MSMS 401 : Bayesian Statistics and Computation

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Problem

Consider a random sample of size n from $\text{Exp}(\lambda)$, λ is the rate parameter. Take a conjugate prior for λ. Obtain the posterior distribution of λ, hence obtain Bayes estimator of λ assuming squared-error loss function.

Solution

Let $x_1, x_2, \dots, x_n \overset{iid}{\sim} f(x \mid \lambda)$ where

$$f(x \mid \lambda) = \lambda e^{-\lambda x}.$$

Consider *a priori* $\lambda \sim \text{Gamma}\,(\text{shape} = \alpha, \text{rate} = \beta)$ *i.e.*

$$\pi(\lambda; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}, \;\; \lambda > 0.$$

The likelihood for λ is

$$f(\mathbf{x} \mid \lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i}$$

$$= \lambda^n \cdot \exp\left\{-\lambda \sum_{i=1}^{n} x_i\right\}$$

We compute the posterior distribution of λ as

$$p(\lambda \mid \mathbf{x}) = \frac{f(\mathbf{x} \mid \lambda) \cdot \pi(\lambda)}{\int\limits_0^\infty f(\mathbf{x} \mid \lambda) \cdot \pi(\lambda)\, d\lambda}$$

$$= \frac{\lambda^n \cdot \exp\left\{-\lambda \sum_{i=1}^{n} x_i\right\} \cdot \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}}{\int\limits_0^\infty \lambda^n \cdot \exp\left\{-\lambda \sum\limits_{i=1}^{n} x_i\right\} \cdot \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}\, d\lambda}$$

$$= \frac{\lambda^{n+\alpha-1} \cdot \exp\left\{-\lambda\left(\sum_{i=1}^{n} x_i + \beta\right)\right\}}{\int\limits_0^{\infty} \lambda^{n+\alpha-1} \cdot \exp\left\{-\lambda\left(\sum_{i=1}^{n} x_i + \beta\right)\right\} d\lambda}$$

$$= \frac{\left(\sum_{i=1}^{n} x_i + \beta\right)^{n+\alpha}}{\Gamma(n+\alpha)} \cdot \lambda^{n-1} \cdot \exp\left\{-\lambda \sum_{i=1}^{n} x_i\right\}.$$

which we identify as the PDF of $\text{Gamma}\left(\text{shape} = \alpha + n, \text{rate} = \beta + \sum_{i=1}^{n} x_i\right)$.

$\therefore$ *a posteriori* $\lambda \mid \mathbf{x} \sim \text{Gamma}\left(\alpha + n, \beta + \sum_{i=1}^{n} x_i\right)$.

Under squared error loss function, Bayes estimate is the posterior mean.

Thus,

$$\hat{\lambda}_{Bayes} = \frac{\alpha + n}{\beta + \sum_{i=1}^{n} x_i}.$$